

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, destination_fd;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    // Open source file for reading
    source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open/create destination file for writing
    destination_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination_fd < 0)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(destination_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source_fd);
            close(destination_fd);
            return 1;
        }
    }

    if (bytes_read < 0)
    {
        perror("Error reading source file");
    }

    // Close both files
    close(source_fd);
    close(destination_fd);

    printf("File copied successfully.\n");
}
```

```
gowtham@DESKTOP-2F03K3T:~$ mkdir 2520090202_oss
gowtham@DESKTOP-2F03K3T:~$ cd 2520090202_oss
gowtham@DESKTOP-2F03K3T:~/2520090202_oss$ mkdir practical.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss$ cd practical.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c$ mkdir practical-2.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c$ cd practical-2.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ nano practical-2.c
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ nano source.txt
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ gcc practical-2.c -o practical-2
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ ./practical-2
File copied successfully.

gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ cat source.txt
Hello OSSP
Linux System Calls

gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ cat destination.txt
Hello OSSP
Linux System Calls

gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ ls -l
total 24
-rw-r--r-- 1 gowtham gowtham  322 Aug  5 05:42 destination.txt
-rwxr-xr-x 1 gowtham gowtham 16176 Aug  5 05:42 practical-2
-rw-r--r-- 1 gowtham gowtham  751 Aug  5 05:42 practical-2.c
-rw-r--r-- 1 gowtham gowtham   33 Aug  5 05:42 source.txt
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c/practical-2.c$ cd ..
gowtham@DESKTOP-2F03K3T:~/2520090202_oss/practical.c$
```